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Computational Biology Analysis

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Evidence 1 | First Analysis

Part 1

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Part I: Scientific context

★ What is a SARS-CoV-2 and what kind of genome does it have?

According to the National Cancer Institute: SARS-CoV-2 belongs to a larger virus family called coronaviruses; these viruses can infect people and some animals. This virus was first known to infect people in 2019 (*NCI Dictionary of Cancer Terms*, n.d.).⁵⁶

Moreover, in a more detailed way SARS-CoV-2 is defined as the “Severe Acute Respiratory Syndrome Coronavirus 2”. It is highly pathogenic and has a single stranded RNA (+ssRNA). SARS-CoV-2 belongs to the order Nidovirales, the family Coronaviridae and the Coronavirinae subfamily ending with the genus Betacoronavirus. It shares an important relationship in evolution with the Severe Acute Respiratory Syndrome Coronavirus.

Its physical structure is spherical, where the diameter of it is from 50 to 200 nanometers. Furthermore, the structure is formed of three structural proteins: the spike glycoprotein, the membrane and the envelope proteins. Internally, it contains the nucleocapsid protein, which binds to the RNA genome to make the formation of a helical ribonucleoprotein structure. The SARS-CoV-2 surface is highly populated with Spike proteins, this structure forms the characteristic crown shape, which facilitates the attachment to the human angiotensin system. Its genome is around 29.9 kb in size, which makes coronaviruses the owners of the largest RNA genomes.

★ What does it mean that the virus has variants?

Viruses variants emerge from the process of its replication and evolution. In scientific terms, a variant can be defined as a version of a virus that mutates one or more times and gets a difference from the original virus first identified in Wuhan, China. It has a proofreading mechanism to reduce errors during replication, with so many infections in the world the virus was replicating a lot and mutations were happening as quickly as the infections.

Variants become prominent when specific mutations get a survival advantage. This is a great example of natural selection, where viruses that are better at replicating, better at attaching to host cells, or better at evading immune defenses are more likely to expand and dominate.

At a molecular level, SARS-CoV-2 evolves through several ways. One common pathway is nucleotide substitution, meaning one base in the genetic code is replaced. If these changes involve amino acids, they can alter the structure or function of virus proteins, potentially improving receptor binding or decreasing the effectiveness of treatments.

★ Mention at least 3 relevant variants and in which regions were they detected.

Since the declaration of the pandemic, the World Health Organization (WHO) has tracked several variants and assigned them Greek letter names to sidestep stigmatization of the locations where they were initially identified. Alpha, Delta and Omicron are three highly significant variants.

The Alpha variant was the first major Variant of concern to get global distress. It was first detected in the United Kingdom in September 2020 and notified to WHO in December 2020. Alpha was distinguished by several mutations, including N501Y in the spike protein, which enhanced its binding affinity. It was associated with an estimated increase in transmissibility of 43%) to 90% compared with the original lineage, and greater risk of severe disease and death.

The Delta variant was first detected in India in late 2020 and became the dominant variant globally by mid-2021. What made it stand out was its high transmissibility, which was about twice as contagious as the original virus. Delta infections were linked to high virality loads, making them more likely to spread. Mutations like L452R and P681R enhanced the virus's ability to enter host cells and provided some immune evasion.

The Omicron variant was reported to the WHO in November 2021 by scientists in South Africa and Botswana. It was a huge evolutionary step because it had an unusually high number of mutations, including more than 30 in the spike protein alone. These mutations helped Omicron avoid much of the immunity built up from prior infections and vaccinations. Explain why it is useful for us to know the virus's genomic/genetic sequences.

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By studying the genomes of viruses scientists can learn in detail about how the virus evolves, spreads and can be restrained.

Daily sequencing of thousands of viral samples means new mutations are identified as they emerge. This process, called genomic surveillance, is a warning system that gives scientists an opportunity to find new variants before they spread and become uncontrollable. It also helps in preparation efforts by providing authorities the ability to anticipate case surges and track the spread of certain variants across regions through tools such as bioinformatics.

Moreover, COVID-19 vaccines train our immune system to identify the spike protein of the virus. But when mutations affect the shape of this protein, the virus can avoid immune system responses. Sequencing data allows researchers to see these changes in detail and update vaccines to keep them effective against circulating variants, just like seasonal flu vaccines are updated. Genomic information can be combined with patient data to allow researchers to explore why some variants cause more severe illness. To understand the relationship between the genotype of a virus and its phenotypes it is vital to predict how new variants may behave and what impact they may have on health.

★ Briefly describe how we can identify variants of a viral sequence

To identify SARS-CoV-2 variants, a series of procedures combined with advanced computational analysis such as bioinformatics.

Some of this process is through sample collection of a clinical specimen (such a q-tip with saliva). The sample is extracted with virus RNA and transcribed into DNA, because DNA is easier to sequence than RNA. The amount of viral material can be low, so it is amplified by PCR.

Furthermore, raw data is further processed through a bioinformatics pipeline after sequencing. First, the data is cleaned and controlled to remove undesired or noisy data and low-quality reads using tools such as fastp or Trimmomatic. The clean sequences are then aligned to a reference genome, usually the original Wuhan-Hu-1 strain, using software such as BWA-mem for short reads or minimap2 for long reads. Finally, all the mutations that are detected are combined to create a consensus sequence, which is the best possible reconstruction of the genome of the virus in that sample.

Part II: Bioinformatic Analysis

Analysis will be presented and found in the following github link:

https://github.com/jjnootfound/Evidence_1_AnalysisOfComputationalBiology.git

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